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Project – HighLevel Design

on

Performance Evaluation of Natural Language Interface for E-commerce Infrastructure

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1.Introduction

1.1 Scope of the document

This document presents the High-Level Design (HLD) for evaluating the performance of a Natural Language Interface (NLI) integrated into an e-commerce platform. The system allows users to interact using natural language queries (text/voice) to search products, track orders, and get recommendations

1.2 Intended Audience

- Developers and SoftwareEngineers
- System Architects
- AI/ML Engineers
- Project Stakeholders
- QA and Performance Testing Teams

1.3 System Overview

The system enables users to:

- Search products using natural language (e.g., "Show me budget smartphones under ₹15,000")
- Get recommendations
- Track orders using conversational input

The architecture integrates:

- Frontend UI
- NLP Engine
- Backend services
- E-commerce database

2. System Design

2.1 Application Design

Modules:
- User Interface (Web/Mobile Chat UI)
- NLP Engine (Intent Recognition + Entity Extraction)
- Query Processor
- Recommendation Engine
- Performance Evaluation Module
- Database

2.2 Process Flow

1. User inputs query (text/voice)
2. Input sent to NLP engine
3.NLP extracts:
- Intent
- Entities

4.Query is converted to structured format
5.Backend processes request
6.Results fetched from database
7.Response returned to user
8.Performance metrics recorded

2.3 Information Flow

User Input → NLP Engine → Intent/Entity Extraction → Query Builder → Backend API → Database → Response → Performance Logger

2.4 Components Design

1. Frontend
- Chat UI
- Voice input (optional)

2. NLP Engine
- Tokenization
- Intent Classification
- Named Entity Recognition (NER)

3. Backend Services
- Product Service
- Order Service
- Recommendation Service

4. Performance Evaluation Module
- Measures latency
- Accuracy tracking
- Logs user interactions

2.5 Key Design Consideration

- Scalability for high user traffic
- Low latency response (< 2 seconds)
- High NLP accuracy
- Fault tolerance
- Modular architecture

2.6 API Catalog

1. Search API

POST /api/search
Request: { query: "budget phones under 15000" }
Response: Product List

2. NLP API

POST /api/nlp
Request: { text: "show me shoes" }
Response: { intent: "search", entities: ["shoes"] }

3.Performance API

POST /api/performance/log
Request: { response_time, accuracy, query }

3. Data Design

3.1 Data Model

Entities:
- User
- Product
- Order
- Query Logs
- Performance Metrics

3.2 Data Access Mechanism

- REST APIs
- ORM (Object Relational Mapping)
- SQL / NoSQL database

3.3 Data Retention Policies

- User queries stored for 6 months
- Performance logs stored for analysis
- Sensitive data encrypted

3.4 Data Migration

- Version-controlled schema updates
- Backup before migration
- Zero downtime deployment

4. Interfaces

Interfaces in the Natural Language Interface (NLI) based e-commerce system define how different components of the system interact and communicate with each other. They ensure smooth data exchange between the user, NLP engine, backend services, and database.

The main interfaces include the User Interface, where users interact through chat or voice input to submit queries. The NLP Interface processes user input and converts it into structured data by identifying intent and entities. The API Interface connects the frontend with backend services using REST APIs for tasks like product search, order tracking, and recommendations. Finally, the Database Interface handles data storage and retrieval, ensuring that user queries fetch accurate results.

- Web UI
- Mobile App
- REST APIs
- NLP Service Integration

4.1 State and Session Management

State and session management in a Natural Language Interface (NLI) for e-commerce ensures that user interactions are handled efficiently and consistently across multiple requests.

In this system, stateless APIs are primarily used, meaning each request from the user is processed independently without storing previous request data on the server. This improves scalability and performance. However, to maintain continuity in conversations (e.g., "show cheaper options" after a search), session management is required.

- Stateless APIs
- JWT-based authentication
- Session stored in cache (Redis)

4.2 Caching

Caching is used to reduce system latency and avoid repeated computation in the NLI-based e-commerce system. Since NLP processing and database queries are expensive operations, caching helps:

- Reduce response time (from seconds → milliseconds)
- Minimize load on NLP models and backend services
- Improve scalability during high traffic

5. Non-Functional Requirements

5.1 Security Aspects

- HTTPS communication
- Authentication & Authorization
- Data encryption
- Protection against injection attacks

5.2 Performance Aspects

- Response time < 2 seconds
- NLP accuracy > 85%
- System availability > 99.5%
- Load handling with auto-scaling

6. References

- NLP Concepts (Intent Recognition, NER)
- E-commerce system architecture best practices
- REST API design guidelines